



Phase 2 -- Backend Development & Configurations

System Architecture & Data Model

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Project Name	Automated Employee Onboarding & Offboarding System
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Document	Data Architecture Design

Architecture Overview

The solution leverages a custom **Employee Lifecycle** data table infrastructure to persist onboarding and offboarding information including Employee ID, Department, Timeline information, and Process Status.

Service Catalog form inputs map to this data table structure to capture employee information during request initiation. **Flow Designer** orchestrates approval workflows and automatically distributes tasks to HR, IT, Facilities, and Security departments.

Role-Based Access Control (RBAC) combined with **Access Control Lists (ACLs)** restrict data access to authorized personnel based on their departmental role. Reporting and dashboard capabilities deliver transparency into request tracking, SLA performance, and comprehensive lifecycle monitoring.

Implementation Elements

Element 1: Data Storage Layer

Goal: Establish custom data storage infrastructure for employee lifecycle records

Activities:

- Engineer custom table infrastructure named **Employee Lifecycle**
 - Define table functional intent for lifecycle data persistence
 - Enable automated record generation during request submissions
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Element 2: Departmental Organization

Goal: Establish organizational structures for departmental task assignment

Activities:

- Configure user group infrastructure for **HR, IT, Facilities, and Security** departments
 - Assign personnel to respective departmental groups
 - Integrate groups into workflow task distribution logic
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Element 3: Record Generation

Goal: Establish automatic record creation mechanisms for lifecycle tracking

Activities:

- Configure the table infrastructure to generate records automatically
 - Ensure captured records reflect comprehensive employee lifecycle information
 - Establish linkage between portal submissions and database records
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Element 4: Data Field Structure

Goal: Define data schema for comprehensive lifecycle information capture

Data Elements:

- Employee ID
- Employee Full Name
- Organizational Department
- Joining Date (for Onboarding)
- Exit Date (for Offboarding)
- Request Type Classification (Onboarding / Offboarding)
- Request Processing Status

Activities:

- Assign appropriate data type configurations for each field
 - Establish mappings between service portal variables and table fields
 - Validate data storage and retrieval mechanisms
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Expected Outcome

- Structured data storage for employee lifecycle management.
- Automated workflow integration with stored data.
- Secure access to data through role-based permissions.
- Improved visibility through reporting and dashboards.